Ordinal Segmentation

From independent pixel decisions to spatially consistent ordered structures

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ORDER-AI
Learning with Order: Ordinal Representations for Trustworthy Medical AI
https://learningwithorder.github.io/
MICCAI 2026

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THESIS

Accuracy is necessary. Trust requires more.



ORDINAL MODELLING

Does the output respect clinical structure?

UNCERTAINTY + CALIBRATION

Does confidence reflect uncertainty?

INTERPRETABILITY

Can I understand the evidence?

Trust emerges from the interaction of all three axes.

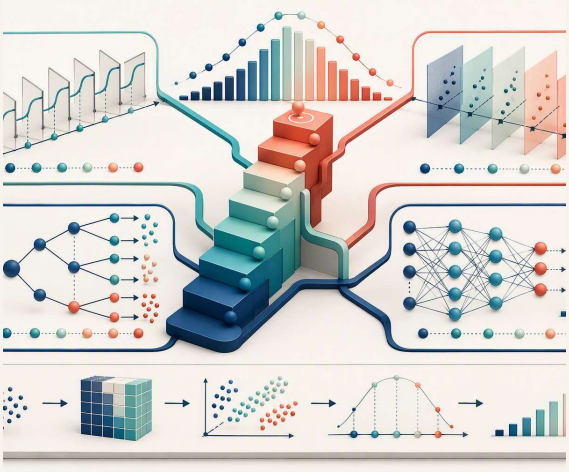
2

ORDINAL CLASSIFICATION

Why ordinary (nominal) classification is not enough

Categories are *naturally* ordered

poor < average < good < very good < excellent



Two consequences

1. **Errors have different severity**
   Predicting an adjacent rank is less damaging than predicting a distant rank.

2. **Order is useful information**
   A model can exploit the ranking to shape boundaries and predictions.

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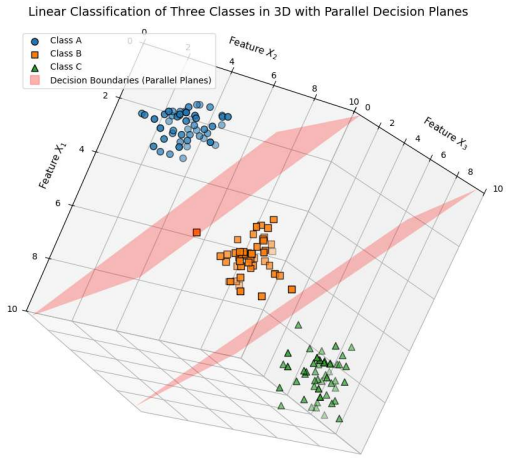
ORDINAL CLASSIFICATION

From ordered output spaces to ordinally consistent models

A decision rule f(x) is ordinal-consistent at a point x_0 if a small perturbation of the input never jumps more than one class:

$$\exists \varepsilon > 0 \text{ such that } \max f(x) - \min f(x) \leq 1$$
$$\text{for all } x \text{ in the } \varepsilon\text{-neighborhood } B\varepsilon(x_0)$$

In plain terms:
decision boundaries in feature space should only separate ordinally adjacent classes.



Small changes in representation or location should not cause large ordinal jumps.

Learning to classify ordinal data: the data replication method, 2007
Classification Models with Global Constraints for Ordinal Data, 2010
oAdaBoost: An AdaBoost variant for Ordinal Classification, 2015

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ORDINAL CLASSIFICATION

Taxonomy (Gutiérrez et al. · IEEE TKDE 2016)



Naïve approaches

Reduce the task to multiclass classification, **cost-sensitive classification** or standard regression.

Binary decompositions

Transform an ordinal target into ordered binary sub-problems, estimated by one or several models.

Threshold models

Assume a latent continuous score partitioned by ordered thresholds.

Examples: SVC / SVR | SVMOP / NNOP | POM / SVOR / GPOR

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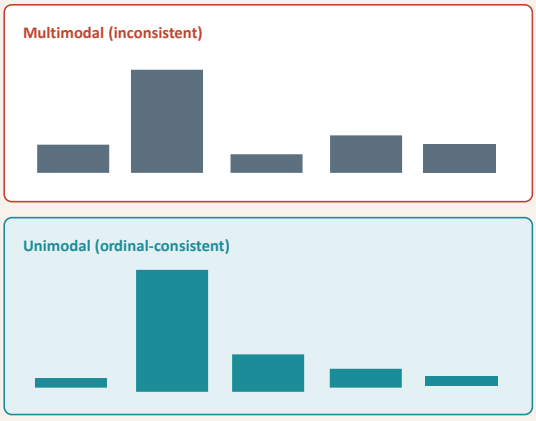
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ORDINAL CLASSIFICATION

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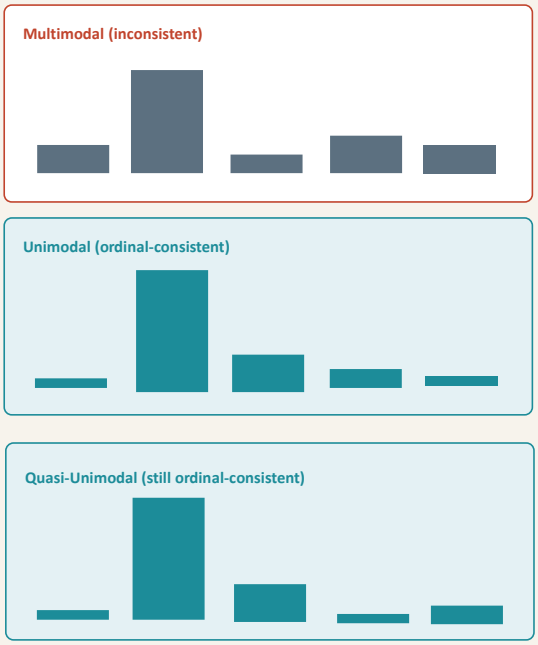
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In plain terms:
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Under mild conditions, unimodality is a sufficient but not necessary condition for ordinally consistent classifier – quasi-unimodality is enough

Quasi-Unimodal Distributions for Ordinal Classification, 2022



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ORDINAL CLASSIFICATION

From ordinal structure to unimodal outputs

UNIMODAL ORDINAL MODELLING

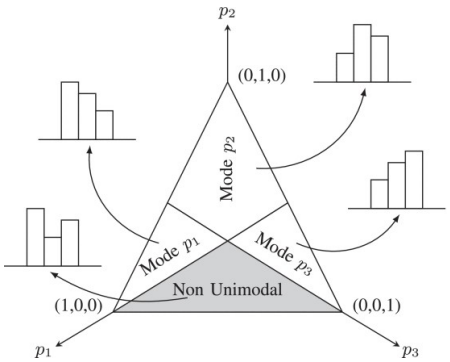
Parametric models
Binomial, Poisson and Gaussian-shaped outputs

Parametric losses
Soft target distributions; unimodal regularization (**KL divergence; Wasserstein distance**)

Non-parametric models
ORD-ACL, VS-SL, CORAL, CORN and **UnimodalNet**

Non-parametric losses
Pairwise monotonic penalties and soft unimodal losses

Unimodal Distributions for Ordinal Regression, Cardoso et al., IEEE TAI 2025



K	3	4	5	6
Unimodal	0.667	0.333	0.133	0.044
Nonunimodal	0.333	0.667	0.867	0.956

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ORDINAL CLASSIFICATION

From ordinal structure to sparse unimodal outputs

PROBABILISTIC ORDINAL MODELLING

Parametric models	Parametric losses	Non-parametric models	Non-parametric losses	THE GAP
Binomial, Poisson and Gaussian-shaped outputs	Soft target distributions and unimodal regularisation	ORD-ACL, VS-SL, CORAL, CORN and UnimodalNet	Pairwise monotonic penalties and soft unimodal losses	Existing approaches provide **unimodality** or **sparsity**

ENABLING FOUNDATIONS

SPARSITY
Sparse probability mappings
Sparsemax assigns exact zeros, but does not enforce unimodality.

PROBABILITIES
Projection onto the simplex
Euclidean projection supplies common thresholding and a tractable Jacobian.

UNIMODALITY
Isotonic regression / PAVA
Increasing and decreasing fits can be joined at a mode.

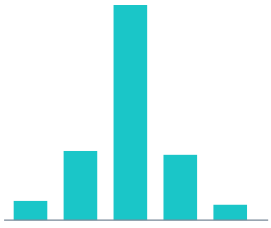
Positioning: a non-parametric output mapping combining unimodal projection with sparse simplex projection.

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ORDINAL CLASSIFICATION

A structured output should be clinically coherent

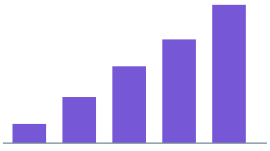
Coherent local ambiguity

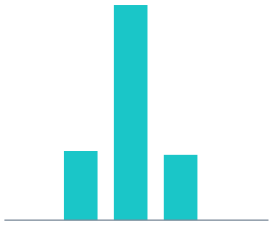


Suspicious multimodality



Coherent severe tendency





Imposing structure is not only regularisation.
It is a statement about plausible predictive behavior.

Sparse outputs improve the **readability** of the posterior by revealing which ranks receive exactly zero probability.

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ORDINAL SEGMENTATION

Why segmentation is central to medical image analysis



Segmentation as the final result

Delineates organs, lesions and anatomical substructures. The mask supports localisation, volumetry, morphology, treatment planning and longitudinal comparison.

Segmentation as an intermediate result

Defines regions of interest for radiomics, classification, registration, image synthesis, report generation and outcome modelling. Errors can propagate to every downstream stage.

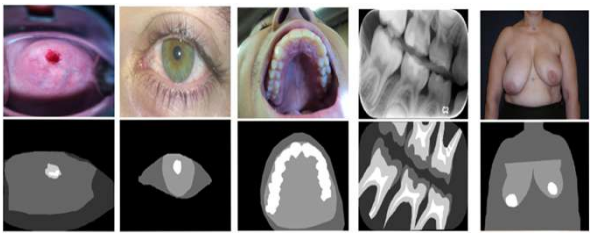
A segmentation mask is both a clinical object and an interface between analysis stages.

Segmentation connects visual evidence to quantitative analysis

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ORDINAL SEGMENTATION

Why ordinality matters in segmentation



Semantic accuracy is not structural validity

Pixel-wise training may permit anatomically or geometrically impossible contacts.

Order is useful domain knowledge

Examples include pupil within iris and lane marking within road. Adjacent regions should respect class order.

The missed opportunity

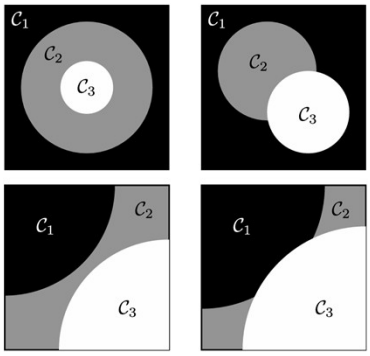
Convolution captures local patterns, but does not explicitly guarantee ordinal neighbour relations.

Cruz, Cristino & Cardoso, IEEE Access, 2025

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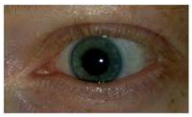
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Cruz, Cristino & Cardoso, IEEE Access, 2025

ORDINAL SEGMENTATION

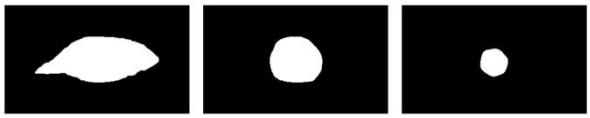
Why ordinality matters in segmentation: initial approaches



(a) Image



(b) Nominal masks



(c) Ordinal masks

Kelwin, Jaime, Ordinal Image Segmentation using Deep Neural Networks

ORDINAL SEGMENTATION

Why representation consistency is not enough

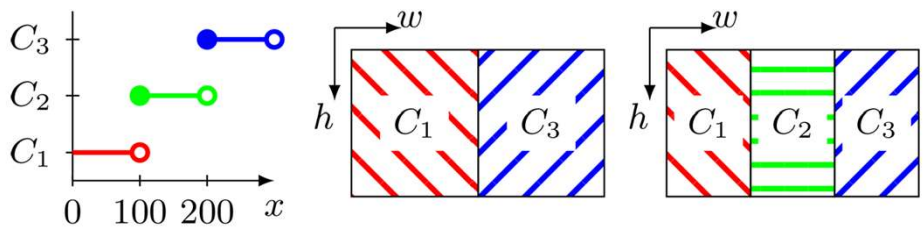


Illustration of possible ordinal (in)consistencies.
(a) Ordinal representation consistency.
(b) Ordinal structure inconsistency.
(c) Ordinal structure consistency.

Cruz, Cristino & Cardoso, IEEE Access, 2025

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ORDINAL SEGMENTATION

Structural consistency: the image is an ordered graph

Pixel grid → graph



valid contacts: $|i - j| \leq 1$



← *penalized: touching, non-adjacent classes*

violation: a non-adjacent class contact

Definition

For every neighbouring pair of pixels (u,v), the predicted labels should satisfy $|f(u) - f(v)| \leq 1$.

What it constrains	Contacts and transitions in the spatial grid, not merely each pixel's posterior.
Why it matters	It prevents anatomically impossible shortcuts, such as background touching an inner structure.
How to learn it	Add a differentiable penalty for probability mass assigned to non-adjacent classes at neighbouring locations.

Representation consistency shapes each pixel's probabilities. Structural consistency shapes which regions may touch.

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ORDINAL SEGMENTATION

A two-axis view of ordinal segmentation

WHERE DOES THE PRIOR ACT?

PIXEL REPRESENTATION

Shape the K-class probability vector at each pixel.

Unimodal · quasi-unimodal · expectation-based

IMAGE STRUCTURE

Shape contacts and transitions between neighbouring regions.

CSNP · CSDT · CSSDF

$$L = L_{seg} + \lambda_{rep} R_{rep} + \lambda_{str} R_{str}$$

accuracy + ordered uncertainty + valid contacts

No architecture change is required. The prior acts during training, so inference cost is unchanged.

Next: losses that implement the two axes

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ORDINAL SEGMENTATION

Loss 1: contact surface using neighbouring pixels

CSNP

$$\mathcal{L}_{CSNP}(\hat{p}_n, \hat{p}_m) = \sum_{v_n \in \mathcal{N}[v_m]} \hat{p}_n^\top C \hat{p}_m$$

$$C_{ij} = \mathrm{ReLU}(|i-j| - 1)$$

Neighbouring pixels are penalised when their probability mass lies on classes separated by more than one ordinal step.

	1	2	3	4
1	0	0	1	2
2	0	0	0	1
3	1	0	0	0
4	2	1	0	0

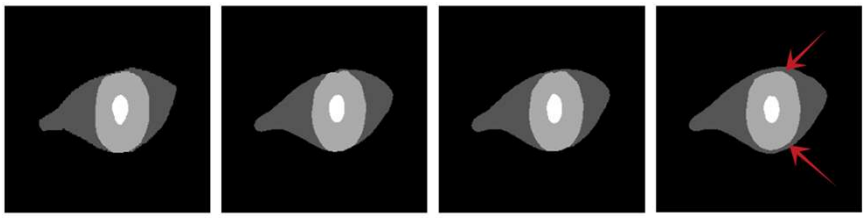
Local, differentiable and architecture-agnostic

Cruz, Cristino & Cardoso, IEEE Access, 2025

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ORDINAL SEGMENTATION

Structural regularisation requires balance



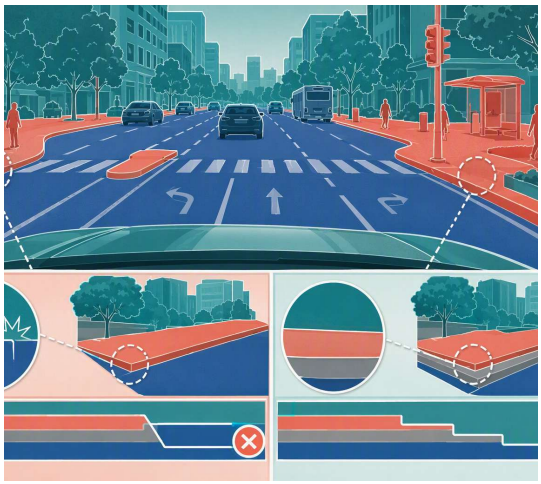
Comparison of the effect of changing the λ coefficient controlling the strength of the CSDT loss term (Mobbio dataset) with (a) Ground-truth, (b) λ = 0.1, (c) λ = 1, (d) λ = 10. The latter (d) is an example of excessive regularization, where the model includes a sclera border (dark gray) between the background (black) and the iris (light gray), which does not exist in the ground truth.

Related direction: richer label structures

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ORDINAL SEGMENTATION

Loss 2: contact surface using distance transforms



Cruz, Cristino & Cardoso, IEEE Access, 2025

CSDT

1. Threshold class probability maps and compute distance transforms.
2. Computes the distance transform of each class mask, then maximizes the distance between the masks of non-adjacent classes. Distance is capped at γ = 10 to avoid exploding, indefinite separation between regions.
3. Penalise geometrically implausible contacts beyond immediate pixel pairs.



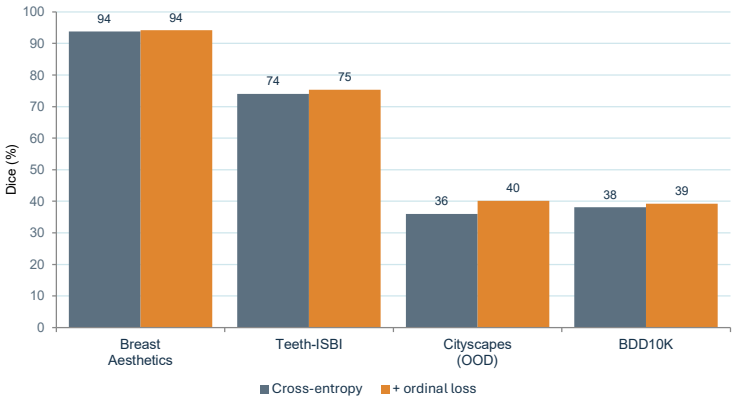
More global geometry. Sees the whole mask, not just a 1-pixel neighborhood — catches farther-apart violations CSNP misses. threshold δ introduces a design choice

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ORDINAL SEGMENTATION

Ordinal regularization improves Dice too

Dice coefficient (%), cross-entropy baseline vs. best ordinal loss — selected datasets



+11.7%

relative Dice gain on out-of-distribution testing (Cityscapes), CE+O2 vs. CE

0

extra inference cost — the losses only apply during training

Evaluated on 5 biomedical datasets and BDD100K / Cityscapes (autonomous driving), with matching gains in the Unimodal-Pixel and Contact-Surface ordinal metrics.

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ORDINAL SEGMENTATION

A broader test: three complementary ordinal losses

Ordinal Semantic Segmentation Applied to Medical and Odontological Images

QUL

QUASI-UNIMODAL

Relax strict shape while preserving ordinal coherence

representation

EXP MSE

EXPECTATION + VARIANCE

Penalise distant expected rank and dispersed posteriors

representation

CSSDF

SPATIAL / SIGNED DISTANCE

Reinforce coherent transitions between adjacent regions

structure

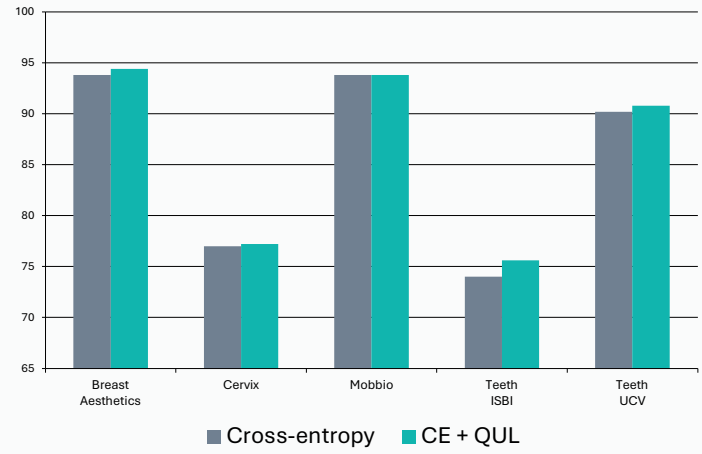
Same U-Net, same datasets, same protocol → isolate what the ordinal loss contributes.

Five medical and odontological datasets · Dice · contact surface · unimodal pixels

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ORDINAL SEGMENTATION

QUL improves Dice while reducing ordinal violations



Mean Dice coefficient (%) · five-fold cross-validation

3 / 5 datasets with the best Dice attained by QUL

Dice Highest with QUL for Breast Aesthetics, Teeth-ISBI and Teeth-UCV.

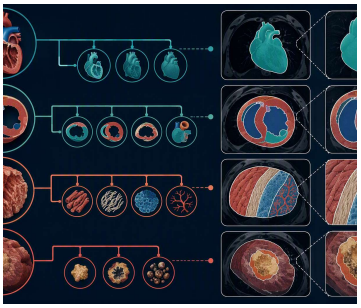
Contact surface QUL reduces non-adjacent contacts versus cross-entropy across all five datasets.

Interpretation A relaxed ordinal prior can improve structure without sacrificing segmentation accuracy.

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RELATED DIRECTIONS

Hierarchical segmentation



$$\mathcal{C}_1 \longrightarrow \mathcal{C}_2 \longrightarrow \mathcal{C}_3 \longrightarrow \mathcal{C}_5$$
$$\mathcal{C}_3 \longrightarrow \mathcal{C}_4$$
$$\mathcal{C}_3 \longrightarrow \mathcal{C}_6 \longrightarrow \mathcal{C}_7$$

Different structure

Hierarchy defines parent-child or part-whole relations. Ordinality defines a ranked sequence. The two structures may coexist, but neither implies the other.

Joint opportunity

A model can enforce both nested labels and valid contacts, for example organ → compartment → lesion while preserving spatial order.

Research question

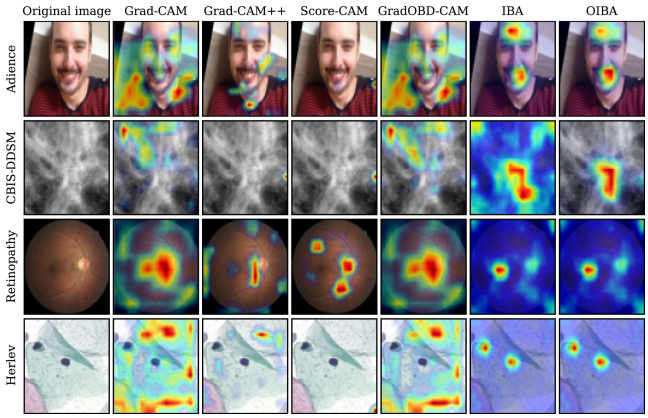
How should losses handle trees, directed acyclic graphs or partial orders rather than a single linear order?

Related direction: richer label structures

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RELATED DIRECTIONS

Explanations for ordinal segmentation



What should be explained?

Not only why a pixel is assigned to class k, but why it is placed between classes k−1 and k+1.

Ordinal explanation objects

Ordered saliency, boundary evidence, contact violations and minimal counterfactual changes along the class order.

Consistency requirement

Explanations should vary smoothly across neighbouring ordered classes and remain faithful to the spatial prediction.

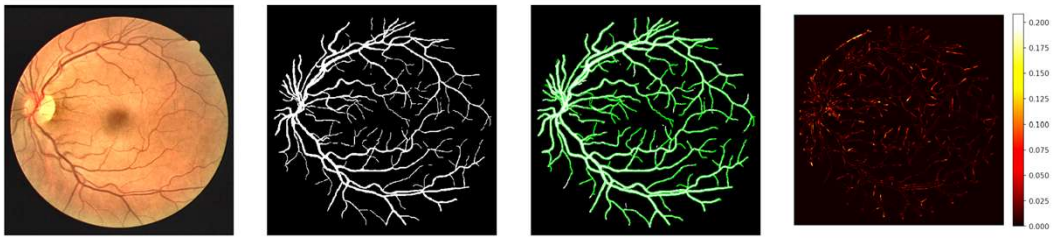
Open direction: explanations that diagnose whether the model learned the intended order.

Related direction: interpretable structured predictions

RELATED DIRECTIONS

Segmentation uncertainty

(Maganti 2026: Rethinking Uncertainty in Segmentation: From Estimation to Decision)



Spatial uncertainty

Highlights uncertain boundaries, ambiguous pixels and regions under distribution shift.

Ordinal uncertainty

Probability mass concentrated on adjacent classes is different from mass split across distant classes. Entropy alone does not encode that distinction.

Clinical use

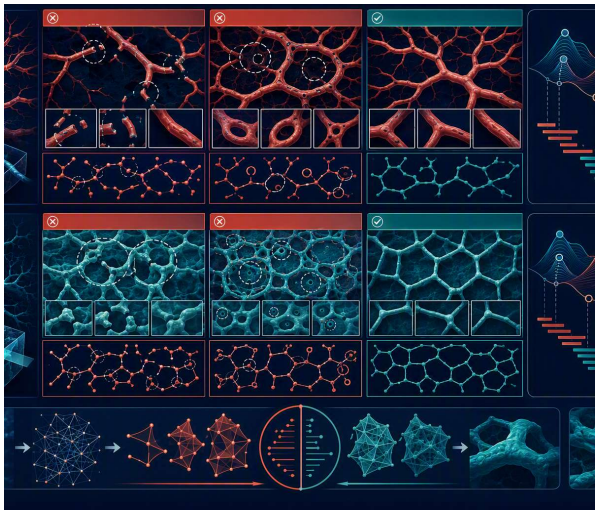
Uncertainty can support quality control, selective review and propagation of confidence to measurements and downstream models.

Open direction: calibration methods and uncertainty scores that respect class order and spatial structure.

Related direction: trustworthy ordinal segmentation

RELATED DIRECTIONS

Topological segmentation



What topology constrains

Connected components, holes, handles and network continuity. Small pixel errors can create major structural failures.

Persistent-homology losses

Compare topological features between the prediction and reference, supplying a differentiable training signal.

Recent extensions

Spatial-aware matching aims to resolve ambiguous feature matching; width-aware formulations also constrain thickness and length.

Ordinality constrains which classes may touch. Topology constrains how regions are connected.

Related direction: global structural guarantees

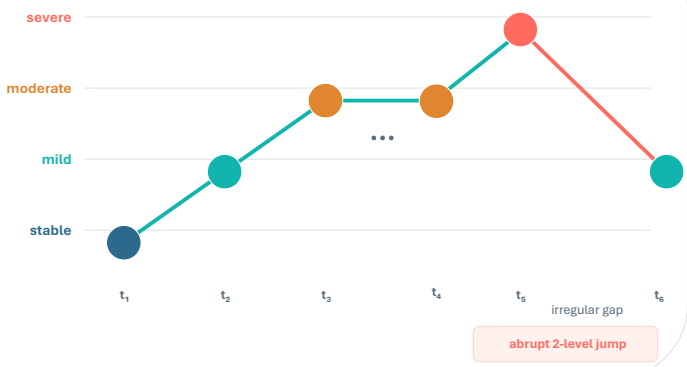
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RELATED DIRECTIONS

Ordinal longitudinal / sequential data / time series

The prediction has two structures: ordered states and ordered time.

Example: disease severity across visits



A good model distinguishes plausible progression, recovery and noise rather than enforcing monotonic change.

CORE CHALLENGES

1. **Two kinds of order**
   Respect rank distances and temporal dependence.
2. **Unequal observation times**
   Visits may be irregular, sparse or missing.
3. **Transitions are asymmetric**
   Progression, recovery and persistence need not be equally likely.
4. **Evaluation is multi-objective**
   Measure rank error, calibration and trajectory coherence.

Open question: what temporal changes are clinically plausible?

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CONCLUSIONS

Take-home messages

Order is not metadata. It is structure that should shape predictions, losses and evaluation.

1 MODEL THE ORDER

Ordinal classification lies between nominal classification and metric regression. Nearby and distant mistakes are not equivalent.

2 SHAPE PLAUSIBLE OUTPUTS

Unimodal and quasi-unimodal posteriors encode coherent ambiguity without treating all classes as unrelated.

3 EXTEND ORDER TO SPACE

In segmentation, pixel-wise consistency is insufficient. Structural consistency also constrains which regions may touch.

4 MEASURE WHAT MATTERS

Complement Dice with ordinal and structural metrics, calibration and, where relevant, temporal coherence.

Design principle **Choose the weakest clinically valid constraint, then verify that it improves coherence without creating artefacts.**

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SELECTED REFERENCES

References and further reading

- Ordinal Semantic Segmentation Applied to Medical and Odontological Images. Mariana Dória Prata Lima, Gilson Antonio Giraldi, Jaime S. Cardoso. https://arxiv.org/abs/2603.26736
- Learning Ordinality in Semantic Segmentation, Ricardo P. M. Cruz, Rafael Cristino, and Jaime S. Cardoso. IEEE Access, 2025
- Ordinal Image Segmentation using Deep Neural Networks. Kelwin Fernandes and Jaime S. Cardoso. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), 2018.
- A Sparse Model for Ordinal Regression. Jaime S. Cardoso. Electronics, 2026.
- Unimodal Distributions for Ordinal Regression. J. S. Cardoso, T. Albuquerque, and R. P. M. Cruz. IEEE Transactions on Artificial Intelligence, 2025.
- A Uniform Performance Index for Ordinal Classification with Imbalanced Classes. Wilson Silva, Joao Ribeiro Pinto, and Jaime S. Cardoso. In Proceedings of the International Joint Conference on Neural Networks (IJCNN), 2018.
- Measuring the performance of ordinal classification. Jaime S. Cardoso and Ricardo Sousa. International Journal of Pattern Recognition and Artificial Intelligence, 2011
- Classification Models with Global Constraints for Ordinal Data, Jaime S. Cardoso and Ricardo Sousa, In Proceedings of The Ninth International Conference on Machine Learning and Applications (ICMLA), 2010
- Learning to classify ordinal data: the data replication method. Jaime S. Cardoso and Joaquim F. Pinto Costa. Journal of Machine Learning Research, 2007.

Disclaimer: Some of the visuals were AI generated

https://jaimecardoso.github.io/

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Thank you, you and… you

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https://jaimecardoso.github.io/